
Graph-Based Stackelberg Model for Adversarial Supply Chain Enforcement

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Abstract

Advanced semiconductor chips underpin modern artificial intelligence systems and have become a strategic driver of military, economic, and technological power. As access to high-end computing increasingly shapes national security outcomes, the United States and allied governments have expanded export controls on advanced computing chips and semiconductor manufacturing equipment. The effectiveness of these controls, however, depends critically on enforcement capacity. Enforcement agencies operate under severe resource constraints relative to the scale and complexity of global semiconductor supply chains. At the same time, smuggling networks can rapidly reconfigure intermediaries, routing, and documentation across permissive jurisdictions. This mismatch creates a structural enforcement challenge. Agencies must commit to inspection and monitoring strategies before observing how adversaries route shipments; adversaries can observe and adapt to these commitments, and enforcement resources are strictly limited relative to the number of potential targets. These features naturally give rise to a leader-follower interaction between regulators and smuggling networks. In this paper, we formalize semiconductor diversion as a Stackelberg security game on a directed supply chain graph. An enforcement agency commits to a probabilistic inspection policy, and an adversary subsequently adapts routing decisions to maximize illicit supplier to end user throughput subject to capacity and budget constraints. The resulting model captures the strategic commitment, adaptive behavior, and resource asymmetries that characterize real-world export control enforcement. By framing enforcement as a network interdiction problem with stochastic detection and constrained evasion, the model provides a foundation for analyzing interdiction effectiveness under adaptive adversarial pressure and limited enforcement capacity.

1 Introduction and Motivation

Semiconductor chips provide the computational foundation for modern artificial intelligence systems. The training and deployment of large-scale models in language, vision, and autonomy rely on advanced accelerator hardware, particularly high-end graphics processing units and specialized AI chips [Semiconductor Industry Association, 2025]. Beyond AI, semiconductors are critical inputs to defense systems, telecommunications infrastructure, medical devices, and energy networks, and have been designated as essential across all sixteen U.S. critical infrastructure sectors [Executive Office of the President, 2026]. As AI capabilities become increasingly central to military and economic power, access to advanced semiconductor technology has emerged as a core national security concern.

Given their strategic importance, governments restrict the sale of the most advanced chips to certain foreign buyers through export control regimes. In the United States, these controls are administered by the Bureau of Industry and Security (BIS) within the Department of Commerce. Since October 2022, BIS has issued multiple rounds of controls targeting advanced computing chips, semiconductor manufacturing equipment, and related technologies, with parallel measures adopted by allied governments in Japan and

the Netherlands [Bureau of Industry and Security, 2022, 2023, Sutter, 2025]. The stated objective of these controls is to prevent adversarial states from acquiring computational capabilities that could accelerate military modernization, weapons development, or mass surveillance.

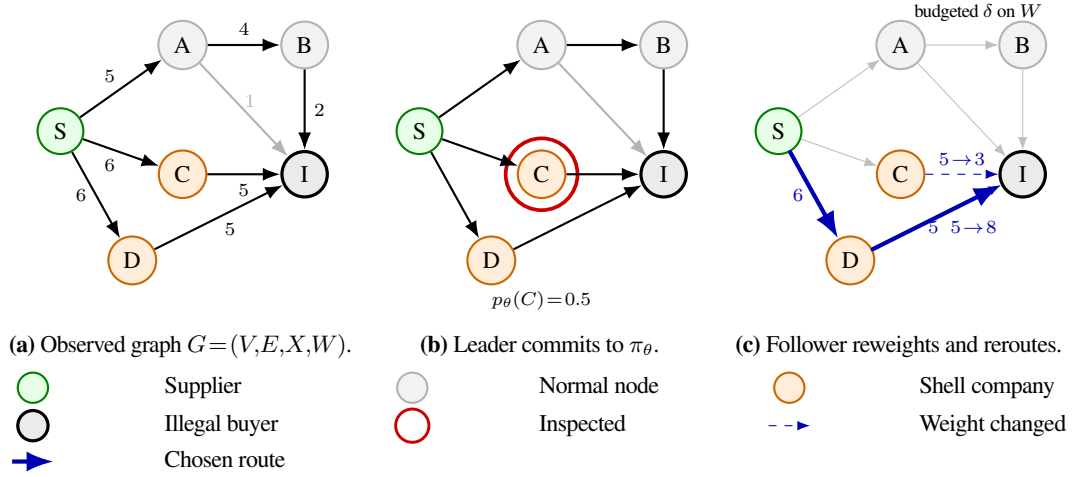


Figure 1: Three-panel schematic of the graph-based Stackelberg enforcement setting: (a) observed directed supply-chain network with baseline edge capacities; (b) the enforcement agency (leader) commits to a probabilistic node inspection policy π_θ ; (c) the smuggling network (follower) adapts by selecting a budgeted edge-weight perturbation δ , yielding $W_f = W + \delta$, and reroutes flow accordingly, illustrating substitution toward an uninspected shell company following partial interdiction.

The effectiveness of export controls, however, depends not only on legal restrictions but on enforcement capacity. Enforcement agencies face a pronounced resource asymmetry. While the global semiconductor trade exceeded \$630 billion in 2024, BIS operated on a budget of approximately \$191 million and maintained a limited number of field agents responsible for monitoring compliance across multiple countries [Sutter, 2025, Grunewald and Fist, 2025]. Individual agents may oversee hundreds of entities and transactions, creating a substantial gap between regulatory scope and enforcement capacity.

Recent enforcement actions illustrate both the scale of semiconductor diversion and the methods used to evade controls. A common strategy involves the creation of shell companies that purchase controlled chips from authorized distributors and reroute them to prohibited end users through transshipment hubs. In 2024, a shell company purchased approximately \$390 million in servers containing restricted Nvidia GPUs before the scheme was detected [Shivakumar et al., 2025]. In another case, a Houston-based firm acquired over \$160 million in advanced GPUs and attempted to route them to China using falsified shipping documents and payments funneled through accounts in Southeast Asia [U.S. Department of Justice, U.S. Attorney’s Office, Southern District of Texas, 2025]. Other networks have used real estate firms or logistics intermediaries as fronts for diversion [U.S. Department of Justice, U.S. Attorney’s Office, Middle District of Florida, 2025]. Syntheses of investigative reporting suggest that smuggled chips may account for a nontrivial share of China’s AI inference capacity acquired in 2024, although precise estimates remain uncertain [Grunewald and Fist, 2025].

Existing enforcement mechanisms struggle to keep pace with these practices. Rule-based screening relies on known red flags and entity lists, but can be circumvented by routing transactions through jurisdictions with limited information sharing or by embedding controlled chips within larger systems that fall outside narrow regulatory definitions [Bureau of Industry and Security, 2025]. Manual investigations are a resource intensive and slow relative to the speed with which shell companies can be created and dissolved. Reactive designation mechanisms such as the Entity List operate only after violations are confirmed, by which point intermediary firms may already have been replaced. Across these approaches, the core challenge is not a lack of detection techniques, but a mismatch between the speed and adaptability of adversarial networks and the limited, pre-committed resources available to enforcement agencies.

This enforcement problem exhibits three structural features that point toward a game-theoretic formulation. First, enforcement agencies must commit to inspection and monitoring strategies before observing how adversaries route shipments, because enforcement posture is partially observable through public actions

and disclosures. Second, adversaries can observe these commitments and adapt their routing strategies in response. Third, enforcement resources are strictly limited relative to the number of potential targets. This structure corresponds to a Stackelberg security game, in which a defender commits to a strategy, and an attacker best responds [Tambe, 2012, Conitzer and Sandholm, 2006]

Main contributions. Motivated by these structural features, we formalize a modeling framework for enforcement resource allocation in adversarial semiconductor supply chains. We represent technology diversion as a Stackelberg security game on a directed supply-chain graph, in which an enforcement agency commits to a probabilistic inspection strategy and an adversary subsequently adapts routing decisions subject to capacity and budget constraints. This formulation captures the structural asymmetries, commitment requirements, and adaptive behavior that characterize real-world export-control enforcement, and provides a foundation for analyzing interdiction effectiveness under resource scarcity.

2 Problem Formulation

Supply chain graphs. We assume the existence of a weighted supply-chain graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, X, W, Y)$, where $\mathcal{V}$ is a set of nodes (e.g., firms or facilities), $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of edges encoding supplier–customer relations, $X \in \mathbb{R}^{|\mathcal{V}| \times d}$ is a matrix of real-valued node attributes, and $W : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0}$ assigns a nonnegative weight to each edge (e.g., transaction volume, dependency strength, or shipment frequency). Each node $v \in \mathcal{V}$ is associated with an attribute vector $\mathbf{x}_v \in \mathbb{R}^d$ from a common feature space. In addition, each node has a categorical label $y_v \in \{0, 1, 2, 3, 4\}$ indicating its role in the supply chain: 0 (supplier), 1 (normal intermediate seller), 2 (shell company), 3 (normal buyer), and 4 (illegal buyer). We collect node labels into a vector $Y \in \{0, 1, 2, 3, 4\}^{|\mathcal{V}|}$ which are then represented in a one-hot manner. To crystallize our problem definition later, we define the supplier set $\mathcal{S} = \{v \in \mathcal{V} : y_v = 0\}$, malicious shell company set $\mathcal{M} = \{v \in \mathcal{V} : y_v = 2\}$, and illegal-buyer set $\mathcal{I} = \{v \in \mathcal{V} : y_v = 4\}$.

When convenient, we equivalently represent the graph $\mathcal{G} = (\mathcal{V}, A, X, Y)$, where $A \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ is a (possibly asymmetric) weighted adjacency matrix with entries $A_{uv} = W(u, v)$ if $(u, v) \in \mathcal{E}$ and $A_{uv} = 0$ otherwise. We assume $\mathcal{G}$ contains no self-loops, i.e., $A_{vv} = 0$ for all $v \in \mathcal{V}$. Unless otherwise stated, edges are directed to reflect the orientation of the underlying supply relation, but the same notation applies to the undirected case by taking A symmetric.

2.1 The Supply Chain Interdiction Problem

Stackelberg security games over a graph. A Stackelberg security game (SSG) models a defender who commits to a (possibly randomized) resource-allocation strategy, after which an attacker observes this commitment and chooses a best-response action. Payoffs depend on whether adversarial activity occurs at *covered* (inspected) versus *uncovered* (uninspected) targets [Tambe, 2012, Conitzer and Sandholm, 2006]. In our setting, targets are nodes of a *known and observed* supply-chain graph $\mathcal{G}$, and defender resources correspond to limited inspections of a node by an enforcement agency. Moreover, the attacker best responds by manipulating edge weights to increase illicit supplier-to-illegal-buyer throughput subject to a fixed budget to avoid trivial and unrealistic scenarios —e.g., setting an edge weight to $+\infty$ between a shell company and the illegal buyer, which acts as a sink.

Leader commitment and inspection coverage. To motivate actual practice, an enforcement agency acts as the leader and commits to an inspection policy that seeks to uncover malicious shell companies that act as proxy gateways to siphon chips into sanctioned markets through sell-offs to illegal buyers. We model this by denoting the leader (defender) commitment to a parametric inspection policy:

$$\pi_\theta : (W, X) \mapsto p_\theta \in \Delta^{n-1}, \quad (1)$$

which outputs an inspection distribution $p_\theta(v)$ over nodes with Δ^{n-1} being the $n - 1$ dimensional probability simplex. This distribution is the leader’s belief based on the graph topology and features in $\mathcal{G}$ that certain nodes $v \in \mathcal{V}$ need an allocation of resources, i.e., interdiction. Given a realized graph, the leader allocates a budget of N inspections by sampling an inspection set $I \subset \mathcal{V}$, by drawing *without replacement* according to π_θ . The resulting *marginal coverage (inclusion) probability* of node v belonging to the inspection set is denoted:

$$c_{\pi_\theta}(v) = \mathbb{P}(v \in I | \pi_\theta), \quad \forall v \in \mathcal{V}. \quad (2)$$

119 Since exactly N distinct nodes are inspected, these inclusion probabilities satisfy the consistency condition

$$\sum_{v \in \mathcal{V}} c_{\pi_\theta}(v) = \sum_{v \in \mathcal{V}} \mathbb{E}[\mathbf{1}(v \in I)] = \mathbb{E}[|I|] = N. \quad (3)$$

120 Intuitively, once the enforcement agency commits to a policy, we observe the outcome—i.e., the interdicted
 121 nodes. While this simplifies the problem for the adversary, in that the adversary has total knowledge of
 122 the interdiction states, it is still a meaningfully difficult optimization problem. Furthermore, it is possible to
 123 also consider the more challenging case of only observing π_θ and not the outcomes, which adds technical
 124 complexity. For the purposes of this paper, we restrict our attention to the simpler and more intuitive case.

125 **Follower attack model.** After observing π_θ , the follower (adversary) perturbs edge weights by choosing
 126 $\delta \in \mathbb{R}^{n \times n}$, yielding $\widetilde{W} = W + \delta$, subject to the feasible attack set

$$\mathcal{D}(W) = \left\{ \delta : \delta_{uv} = 0 \text{ if } (u, v) \notin \mathcal{E}; |\delta_{uv}| \leq \Gamma \forall (u, v) \in \widetilde{\mathcal{E}}; \sum_{(u, v) \in \widetilde{\mathcal{E}}} |\delta_{uv}| \leq B; W + \delta \geq 0 \right\}. \quad (4)$$

127 Here $\widetilde{\mathcal{E}} := \bigcup_{v \in \mathcal{M}} \mathcal{N}_1(v)$ is the edge set belonging to all nodes that fall in the one-hop neighborhood, $\mathcal{N}_1$,
 128 of any shell company $v \in \mathcal{M}$. In addition, Γ bounds the per-edge perturbation magnitude, and B is the total
 129 attack budget. Given capacities $\widetilde{W} \in \mathbb{R}_{\geq 0}^{n \times n}$, define the feasible set of flows on the original edge set $\mathcal{E}$ as

$$\mathcal{F}(\widetilde{W}) = \left\{ f \in \mathbb{R}_{\geq 0}^{n \times n} : f_{uv} = 0 \text{ if } (u, v) \notin \mathcal{E}; \sum_{u: (u, v) \in \mathcal{E}} f_{uv} = \sum_{w: (v, w) \in \mathcal{E}} f_{vw} \forall v \in \mathcal{V} \setminus (\mathcal{S} \cup \mathcal{I}) \right\}. \quad (5)$$

130 That is, flow is capacity-feasible and conserved at all *non-terminal* nodes; supplier nodes $\mathcal{S}$ may inject
 131 net flow and illegal-buyer nodes $\mathcal{I}$ may absorb net flow. Without loss of generality, we take the flow f
 132 to be the edge weight. and for any feasible f , we define the total flow delivered into a particular illegal
 133 buyer $i \in \mathcal{I}$ as its inflow

$$F_i(f) = \sum_{u: (u, i) \in \mathcal{E}} f_{ui}. \quad (6)$$

134 Next, we introduce and define the total supplier-to-illegal-buyer throughput as the multi-source, multi-sink
 135 max-flow value over our prescribed supply-chain graph $\mathcal{G}$,

$$F_{\mathcal{S} \rightarrow \mathcal{I}}(\widetilde{W}) = \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} F_i(f) = \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} \sum_{u: (u, i) \in \mathcal{E}} f_{ui}. \quad (7)$$

136 **Detection model and utilities.** We adopt a sink-interdiction model: if an illegal buyer $i \in \mathcal{I}$ is inspected,
 137 illicit throughput terminating at i is seized/blocked. In addition, a shell company can also be interdicted,
 138 resulting in all the flow at that node being set to zero. Under without-replacement inspection, the probability
 139 that i is not inspected is $1 - c_{\pi_\theta}(i)$. Thus, for a fixed π_θ , the attacker's expected *successful* illicit throughput
 140 under a flow f is then simply stated as:

$$\sum_{i \in \mathcal{I}} (1 - c_{\pi_\theta}(i)) F_i(f). \quad (8)$$

141 The attacker chooses δ (hence $\widetilde{W}$) to maximize expected successful illicit throughput, subject to optimal
 142 routing of flow. Including an optional manipulation-effort penalty $\kappa \geq 0$, the follower utility is

$$U_A(\theta, \delta) = \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} (1 - c_{\pi_\theta}(i)) F_i(f) - \kappa \sum_{(u, v) \in \mathcal{E}} |\delta_{uv}|. \quad (9)$$

143 In contrast, the enforcement agencies, ie, defenders, seek to minimize successful illicit throughput while
 144 accounting for inspection costs. A canonical defender utility is:

$$U_D(\theta, \delta) = - \max_{f \in \mathcal{F}(\widetilde{W})} \sum_{i \in \mathcal{I}} (1 - c_{\pi_\theta}(i)) F_i(f) - \lambda \sum_{v \in \mathcal{V}} c_v(p_\theta) C_v, \quad (10)$$

145 where $C_v \geq 0$ is a per-node inspection burden and $\lambda \geq 0$ trades off interdiction vs. cost. Alternative utilities
 146 can also be considered, such as maximizing the interdiction set I , containing as many shell company and
 147 illegal buyer nodes. However, our framing in eq. (10) poses a natural foil to the attacker's utilities.

2.2 Stackelberg equilibrium

Given the definition of the Stackelberg game along with the payoffs of each player, it is natural to consider the notion of equilibria. We can now pose this question explicitly by stating the objective for each player’s best response. More precisely, given the leader’s commitment π_θ , the follower’s best response is

$$\delta^*(\theta) \in \arg \max_{\delta \in \mathcal{D}(W)} U_A(\theta, \delta). \quad (11)$$

The leader then chooses θ anticipating this best response:

$$\theta^* \in \arg \max_{\theta} U_D(\theta, \delta^*(\theta)). \quad (12)$$

We refer to this as a *Graph Stackelberg Security Game with Budgeted Edge-Weight Attacks* and a *multi-source, multi-sink max-flow* adversarial objective.

3 Related Work

Security games and Stackelberg models. Stackelberg security games model a defender who commits to a resource allocation strategy and an attacker who best-responds after observing that commitment [Tambe, 2012, Conitzer and Sandholm, 2006]. This framework has supported deployed decision aids in real-world protection settings, demonstrating that game-theoretic allocation can improve outcomes under scarce resources [Fang et al., 2016]. Our work uses the Stackelberg structure but focuses on supply-chain diversion where the attacker adapts by shifting flow through an evolving network.

Network interdiction and bilevel optimization. Network interdiction problems formalize a leader who disables or constrains parts of a network to reduce an adversary’s objective, such as maximum flow [Wood, 1993]. The literature includes algorithmic developments and surveys emphasizing the bilevel nature of interdiction and the computational challenges it creates [Smith and Song, 2020]. Extensions study multi-objective and cost tradeoffs in flow interdiction [Royset and Wood, 2007]. Our formulation aligns with this tradition but emphasizes learning policies that can generalize across changing graphs.

Learning for interdiction and security domains. Recent work explores machine learning and reinforcement learning as scalable alternatives or complements to exact optimization for interdiction-style problems [Baycik, 2022, Huang et al., 2023]. In security-game settings, deep reinforcement learning has been used to compute policies in structured patrol and protection domains under uncertainty and partial observability [Wang et al., 2019].

4 Limitations and Future Work

The framework in this paper is intentionally stylized and subject to several limitations. We assume full observability of the supply-chain network, whereas in practice enforcement agencies operate under substantial uncertainty regarding hidden shell companies, relationships, and routing behavior. Key parameters such as detection probabilities, edge capacities, and adversarial manipulation budgets are estimated rather than empirically calibrated, reflecting limited availability of ground-truth data on successful diversion.

The current formulation evaluates enforcement and adversarial response within a single episode on a fixed graph instance. While the Stackelberg equilibrium conceptually assumes that the leader’s policy is learned or optimized over repeated interactions, we do not model this learning process explicitly. Real-world export control enforcement is inherently repeated, with agencies updating inspection priorities over time and smuggling networks adapting in response. Future work could extend the model to a multi-episode setting, enabling policies to be trained and evaluated across repeated interactions and a distribution of network instances, and allowing analysis of learning dynamics and generalization beyond a single realization.

Similarly, the supply-chain graph is treated as static. In practice, shell companies are created, dissolved, and re-linked in response to enforcement pressure, with routing patterns evolving accordingly. An important extension is to model the supply chain as a dynamic graph with endogenous node and edge evolution, including shell-company birth and death processes, time-varying capacities, and adaptive rewiring. Such dynamics would better capture the lag between reactive designation mechanisms and rapid network reconfiguration.

Finally, scaling the framework to larger or evolving networks motivates learning-based solution methods for settings where exact bilevel optimization becomes intractable. Reinforcement learning approaches

194 combined with graph-based representations offer a natural direction for learning inspection policies that
195 adapt over time while remaining grounded in the Stackelberg structure formalized here.

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